KEDGE BUSINESS SCHOOL · QM1 · 2026–2027

# Worksheet 1

Presenting and organising data

**Fully worked solutions**

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## 1. Core applications

Master the essential definitions and calculations from the course.

Core application 10 min

### Exercise 1 – Population, sample and variable types

A store wants to study all customers who made a purchase in September. It surveys a sample of 80 customers and records:

| Variable | Example |
|---|---|
| Payment method | card, cash, mobile |
| Number of items purchased | 1, 2, 3, … |
| Time spent in store | minutes |
| Satisfaction level | very dissatisfied to very satisfied |

1. Identify the target population, the sample and the statistical unit.
2. Classify each variable as nominal qualitative, ordinal qualitative, discrete quantitative or continuous quantitative.
3. Choose an appropriate graph for each variable.
4. For which variables do cumulative frequencies make sense? Explain.
5. Explain why a large sample is not necessarily representative.

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### Exercise 1 – Detailed solution

Target population: all customers who made a purchase in September. Sample: the 80 surveyed customers. Statistical unit: one customer.

| Variable | Type | Suitable graph |
|---|---|---|
| Payment method | nominal qualitative | bar chart / pie chart |
| Number of items | discrete quantitative | stick/bar diagram |
| Time in store | continuous quantitative | histogram |
| Satisfaction | ordinal qualitative | ordered bars |

Cumulative frequencies are meaningful for ordered variables: number of items, time and satisfaction. They are not meaningful for payment method because nominal categories have no natural order.

A large sample may still be biased if selection is not representative. For example, a voluntary survey may attract more highly satisfied or highly engaged customers.

Core application 20 min

### Exercise 2 – Discrete distribution

| $x_i$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $n_i$ | 3 | 7 | 12 | 10 | 6 | 2 |

1. Check the total count $N$.
2. Calculate relative frequencies $f_i = n_i/N$ and increasing cumulative frequencies.
3. Draw a stick/bar diagram.
4. Construct the cumulative distribution function $F(x) = P(X \le x)$.
5. Find the proportion of customers who placed at most three orders.

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### Exercise 2 – Detailed solution

$$N = 3 + 7 + 12 + 10 + 6 + 2 = 40.$$

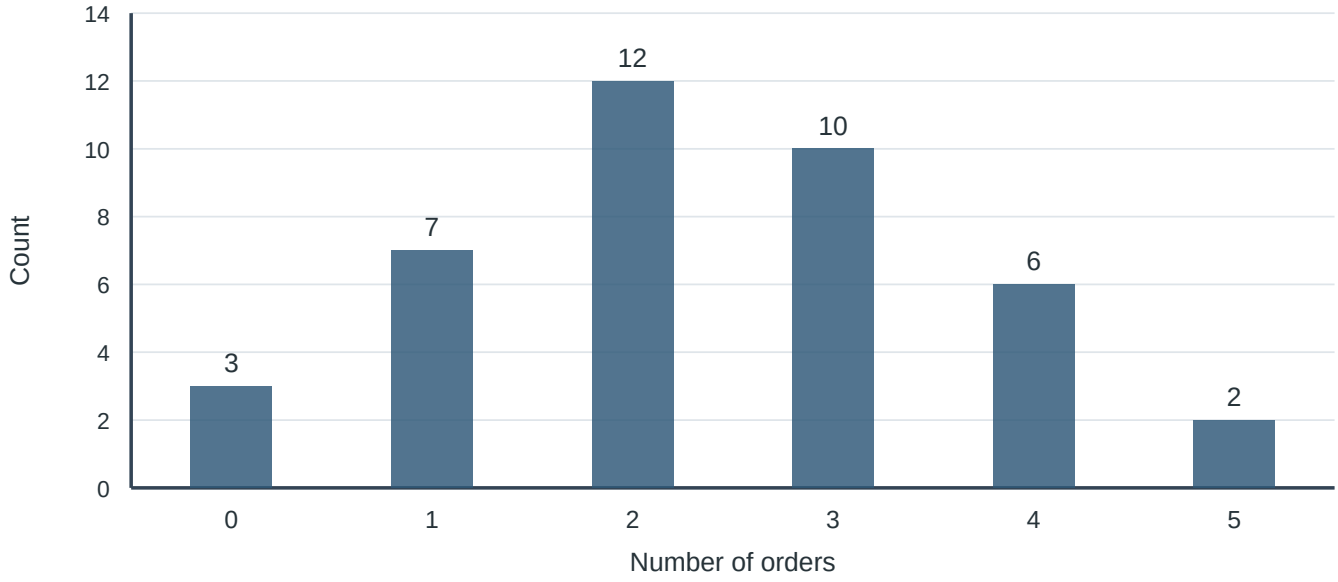
| $x_i$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $n_i$ | 3 | 7 | 12 | 10 | 6 | 2 |
| $f_i$ | 0.075 | 0.175 | 0.300 | 0.250 | 0.150 | 0.050 |
| $F_i$ | 0.075 | 0.250 | 0.550 | 0.800 | 0.950 | 1.000 |

For example:

$$f_2 = \frac{12}{40} = 0.30 = 30\%.$$

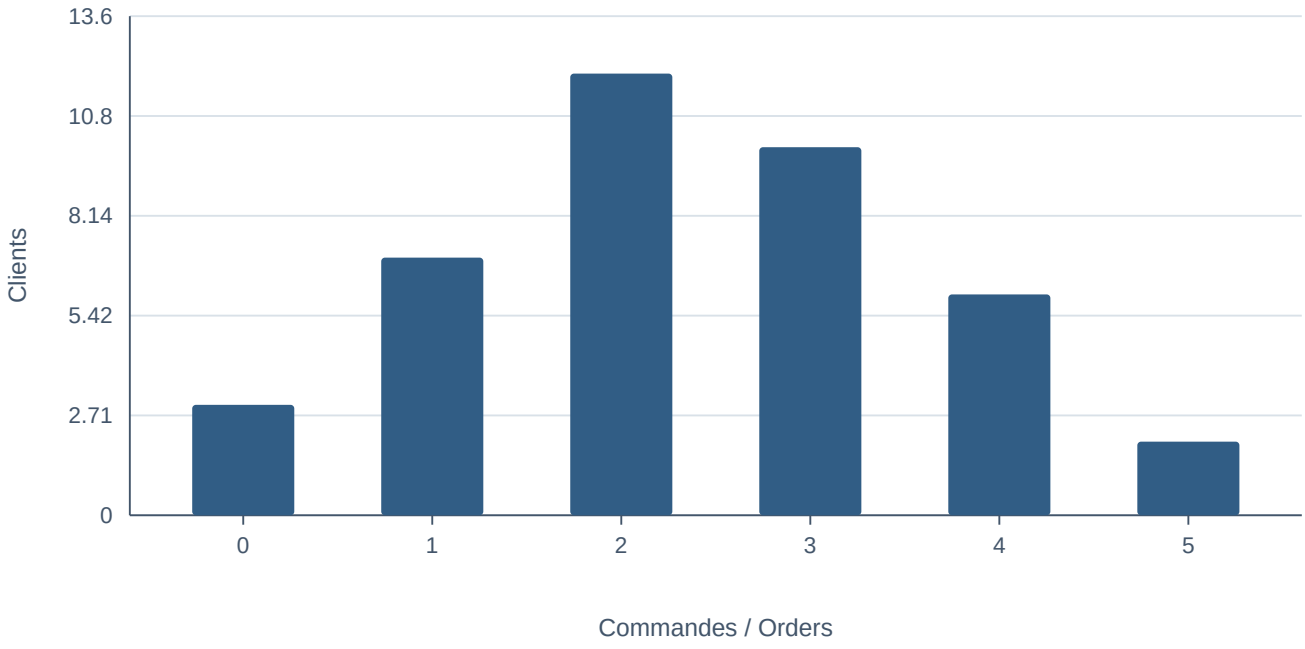
The proportion with at most three orders is:

$$F(3) = \frac{3 + 7 + 12 + 10}{40} = \frac{32}{40} = 0.80 = 80\%.$$

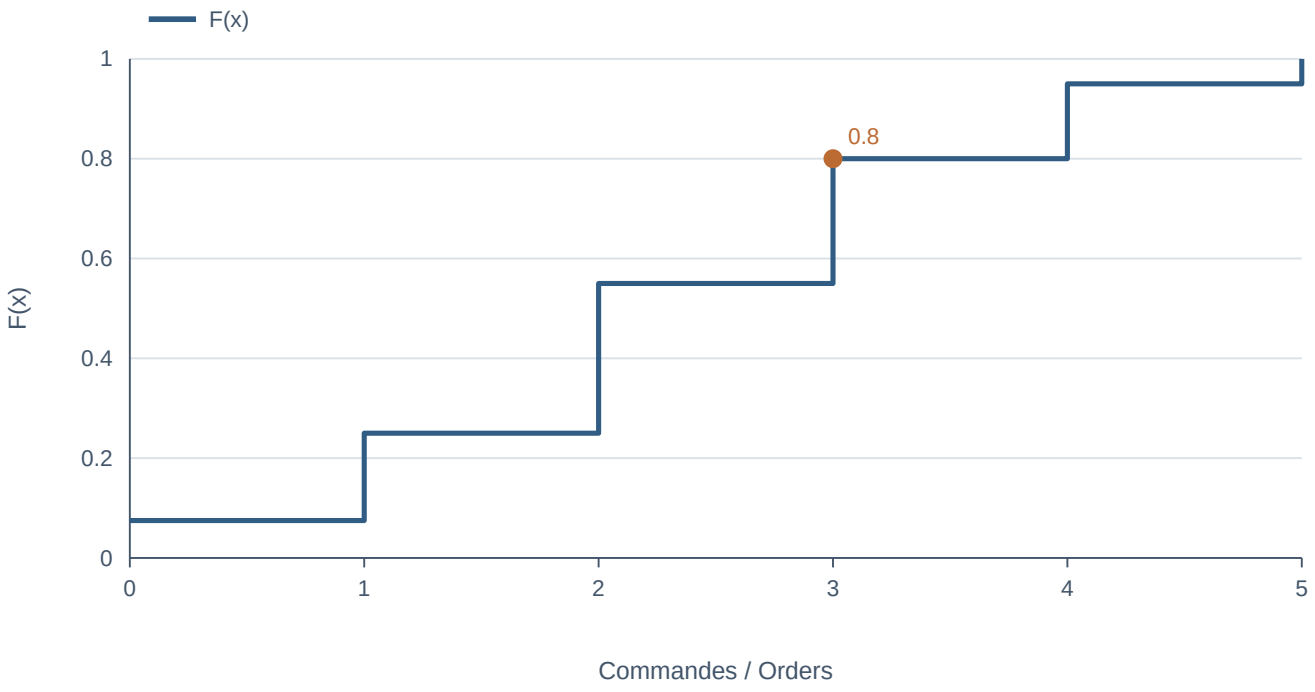



Bar chart: number of orders

Charts from the counts: the cumulative distribution reaches 80% at x=3.




Nombre de commandes / Orders




Fonction de répartition / Empirical CDF

**Complete cumulative table**

| x | n | f=n/40 | ECC | ECD | FCC | FCD |
|---|---|---|---|---|---|---|
| 0 | 3 | 3/40=.075 | 3 | 40 | .075 | 1.000 |
| 1 | 7 | 7/40=.175 | 10 | 37 | .250 | .925 |
| 2 | 12 | 12/40=.300 | 22 | 30 | .550 | .750 |
| 3 | 10 | 10/40=.250 | 32 | 18 | .800 | .450 |
| 4 | 6 | 6/40=.150 | 38 | 8 | .950 | .200 |
| 5 | 2 | 2/40=.050 | 40 | 2 | 1.000 | .050 |

At x=3, ascending count=3+7+12+10=32 and descending count=10+6+2=18. Ascending cumulative frequencies measure $P(X \le x)$, descending ones $P(X \ge x)$.

Core application 15 min

## Extra 1 — Cross-tabulation and conditional percentages

Among 120 students, 50 use public transport, including 30 women. Of the 70 other students, 35 are women.

1. Build the gender × transport table.
2. Compute the public-transport rate within each gender.
3. State whether an association appears present.

Conditional percentages use the total of the conditioning group as denominator.

### Extra 1 — Cross-tabulation and conditional percentages

### Detailed solution

The table is: women 30 public and 35 other; men 20 public and 35 other. Totals are 65 women and 55 men. Conditional rates are 30/65 = 46.2% for women and 20/55 = 36.4% for men.

The sample shows a 9.8-point difference. This is descriptive evidence of association, not proof of a population difference or causal effect.

## 2. Assessment and exam practice

Practise independently with assessment and final-exam style questions.

Assessment / exam practice  40 min

### Exercise 3 – Continuous variable with unequal class widths

| Class | $[10,14[$ | $[14,18[$ | $[18,26[$ | $[26,30[$ | $[30,35[$ | $[35,40[$ | $[40,45[$ | $[45,50[$ | $[50,55[$ |
|---|---|---|---|---|---|---|---|---|---|
| $n_i$ | 4 | 12 | 32 | 32 | 60 | 35 | 25 | 5 | 5 |

1. For each class, calculate width $a_i$, relative frequency $f_i$ and count density $d_i = n_i/a_i$.
2. Draw the correct histogram and explain why raw counts cannot be used as heights when class widths differ.
3. Calculate increasing and decreasing cumulative counts and the corresponding cumulative relative frequencies.
4. Draw the cumulative curves.
5. Find the number and proportion willing to pay less than 35 euros, then at least 35 euros.

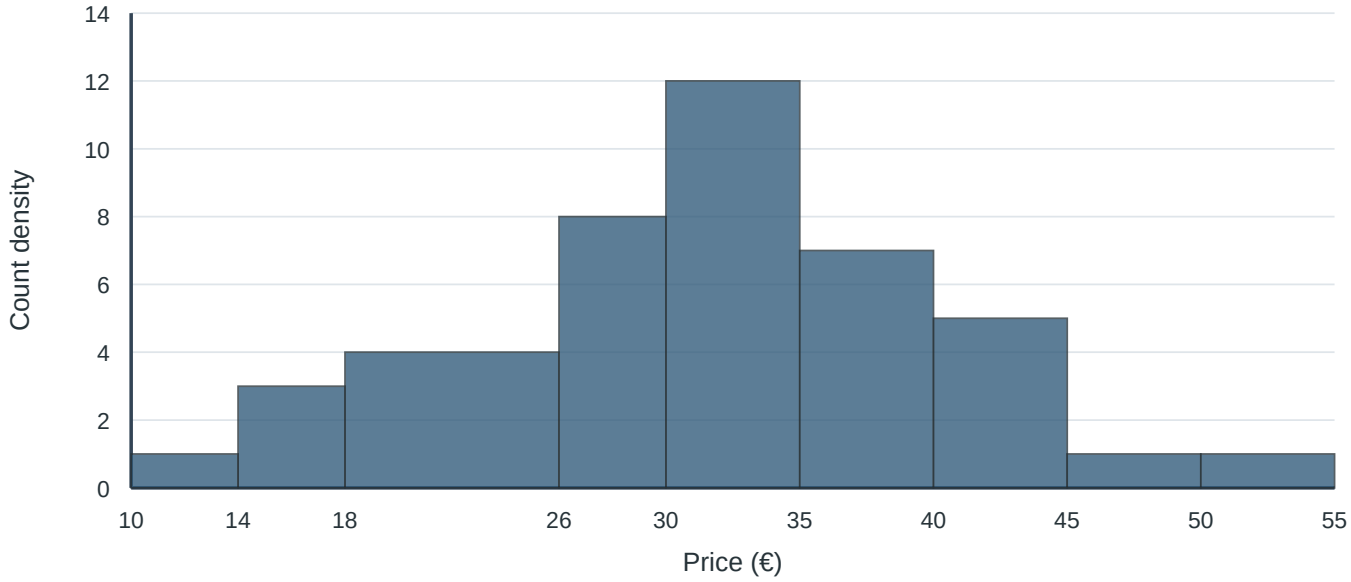
### Exercise 3 – Detailed solution

| Class | $[10,14[$ | $[14,18[$ | $[18,26[$ | $[26,30[$ | $[30,35[$ | $[35,40[$ | $[40,45[$ | $[45,50[$ | $[50,55[$ |
|---|---|---|---|---|---|---|---|---|---|
| $n_i$ | 4 | 12 | 32 | 32 | 60 | 35 | 25 | 5 | 5 |
| $a_i$ | 4 | 4 | 8 | 4 | 5 | 5 | 5 | 5 | 5 |
| $f_i(\%)$ | 1.90 | 5.71 | 15.24 | 15.24 | 28.57 | 16.67 | 11.90 | 2.38 | 2.38 |
| $d_i$ | 1 | 3 | 4 | 8 | 12 | 7 | 5 | 1 | 1 |
| ICC | 4 | 16 | 48 | 80 | 140 | 175 | 200 | 205 | 210 |
| DCC | 210 | 206 | 194 | 162 | 130 | 70 | 35 | 10 | 5 |

For example:

$$f_{[18,26[} = \frac{32}{210} \approx 0.1524 = 15.24\%, \qquad d_{[18,26[} = \frac{32}{8} = 4.$$

With unequal class widths, rectangle area must represent the count, so density is used as height.




Histogram with unequal class widths

Less than 35 euros:

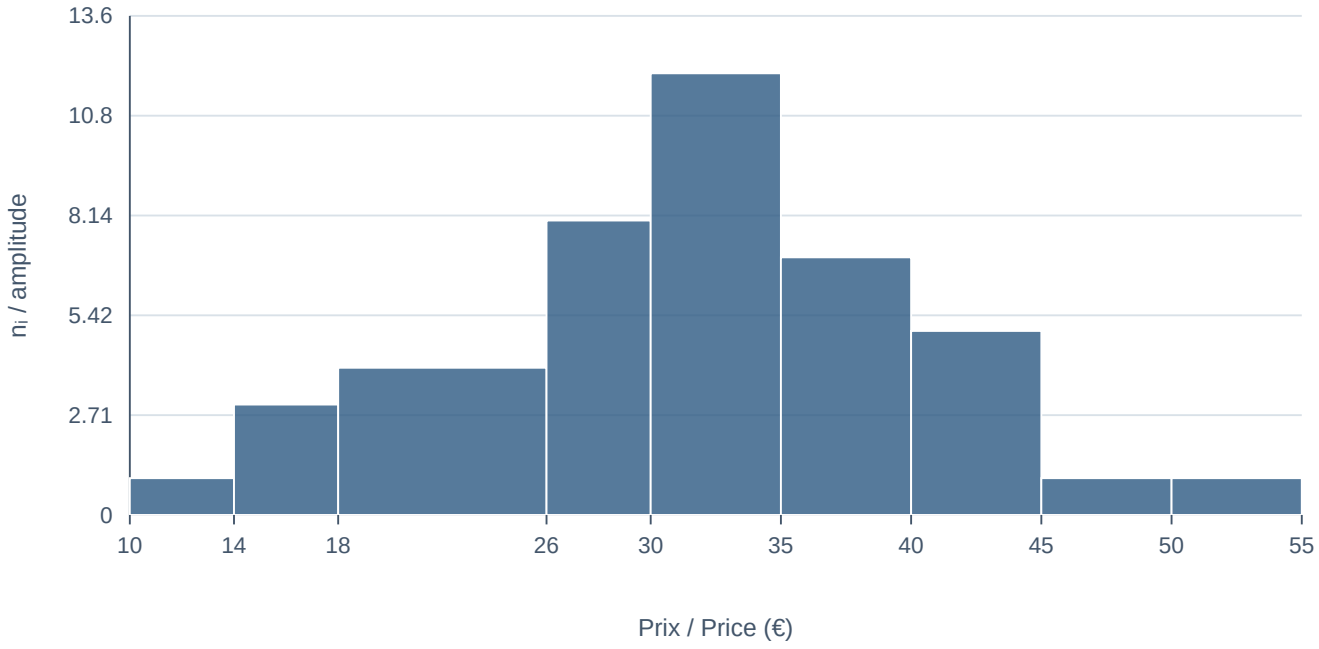
$$140, \qquad \frac{140}{210} = 66.67\%.$$

At least 35 euros:

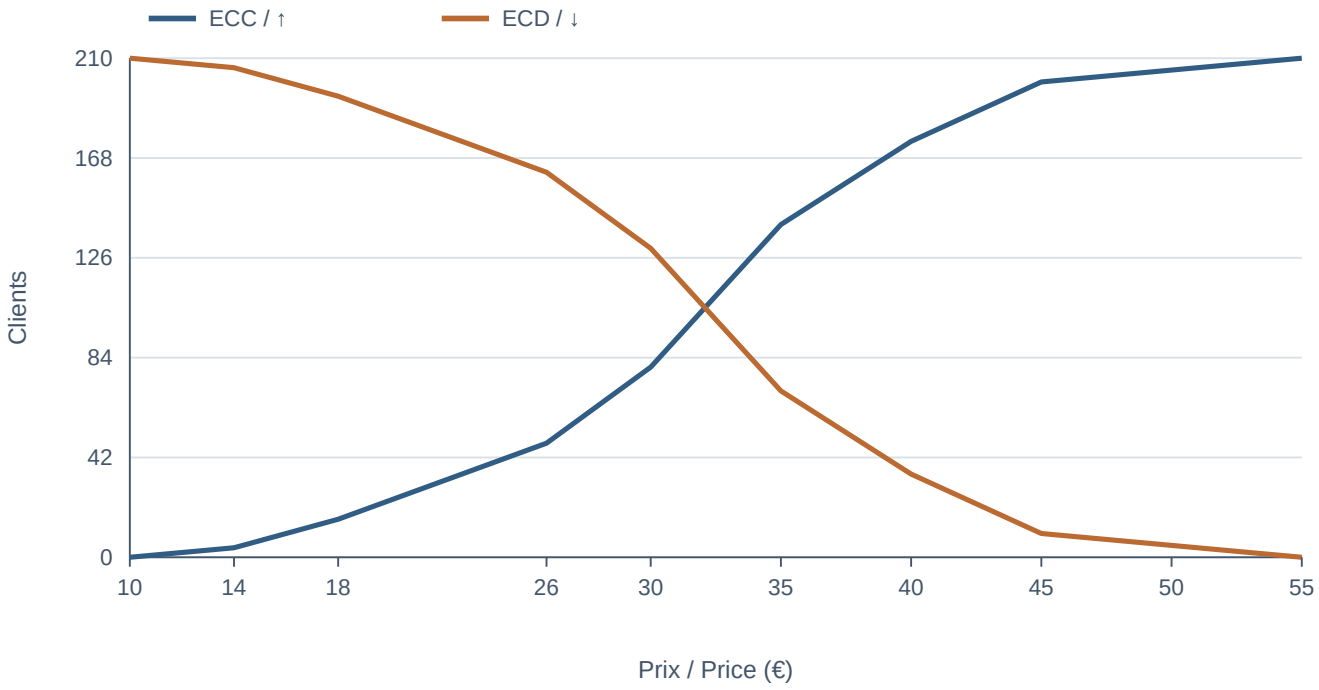
$$70, \qquad \frac{70}{210} = 33.33\%.$$

Complete table: $f_i = n_i/210$; cumulative relative frequencies are ECC/210 and ECD/210. Cumulative plots use class boundaries.

| Classe | $a_i$ | $n_i$ | $f_i$ (%) | $d_i$ | ECC | ECD | FCC (%) | FCD (%) |
|---|---|---|---|---|---|---|---|---|
| [10,14[ | 4 | 4 | 1.90 | 1 | 4 | 210 | 1.90 | 100.00 |
| [14,18[ | 4 | 12 | 5.71 | 3 | 16 | 206 | 7.62 | 98.10 |
| [18,26[ | 8 | 32 | 15.24 | 4 | 48 | 194 | 22.86 | 92.38 |
| [26,30[ | 4 | 32 | 15.24 | 8 | 80 | 162 | 38.10 | 77.14 |
| [30,35[ | 5 | 60 | 28.57 | 12 | 140 | 130 | 66.67 | 61.90 |
| [35,40[ | 5 | 35 | 16.67 | 7 | 175 | 70 | 83.33 | 33.33 |
| [40,45[ | 5 | 25 | 11.90 | 5 | 200 | 35 | 95.24 | 16.67 |
| [45,50[ | 5 | 5 | 2.38 | 1 | 205 | 10 | 97.62 | 4.76 |
| [50,55[ | 5 | 5 | 2.38 | 1 | 210 | 5 | 100.00 | 2.38 |
| TOTAL | — | 210 | 100 | — | 210 | — | 100 | — |




Histogramme des densités / Density histogram




Effectifs cumulés / Cumulative counts

Assessment / exam practice 35 min

## Extra 3 — Continuous-assessment case: willingness to pay

**Case study — 12 marks.** A food retailer asks 200 customers for the maximum price they would pay for a sustainable lunch box.

| **Maximum price (€)** | **[20,25[** | **[25,30[** | **[30,40[** | **[40,50[** | **[50,60[** | **[60,80[** |
|---|---|---|---|---|---|---|
| **Customers** | 10 | 20 | 50 | 60 | 40 | 20 |

1. Identify the population, statistical unit, variable and variable type. *(2 marks)*
2. Calculate class widths, relative frequencies and frequency densities. *(3 marks)*
3. Explain how to construct the correct histogram and identify the modal class. *(2 marks)*
4. Calculate ascending and descending cumulative counts. Give the coordinates needed to draw both cumulative curves. *(3 marks)*
5. Find the highest price that at least 60% of surveyed customers would be willing to pay. Explain the percentile used. *(2 marks)*

Unequal-width classes require frequency density. For the pricing decision, translate “at least 60% can pay” into an ascending cumulative percentage.

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## Extra 3 — Continuous-assessment case: willingness to pay

### Detailed solution and marking logic

**1.** Population: the 200 surveyed customers; unit: one surveyed customer; variable: maximum willingness to pay in euros; type: quantitative continuous.

| Class | Width | Count | Frequency | Density (count/width) | Ascending count |
|---|---|---|---|---|---|
| [20,25[ | 5 | 10 | 5% | 2 | 10 |
| [25,30[ | 5 | 20 | 10% | 4 | 30 |
| [30,40[ | 10 | 50 | 25% | 5 | 80 |
| [40,50[ | 10 | 60 | 30% | 6 | 140 |
| [50,60[ | 10 | 40 | 20% | 4 | 180 |
| [60,80[ | 20 | 20 | 10% | 1 | 200 |

**2–3.** Because widths differ, histogram heights must be densities, not raw counts. The largest density is 6, so the modal class is [40,50[.

**4.** The ascending curve passes through (20,0), (25,10), (30,30), (40,80), (50,140), (60,180), (80,200). The descending curve uses (20,200), (25,190), (30,170), (40,120), (50,60), (60,20), (80,0).

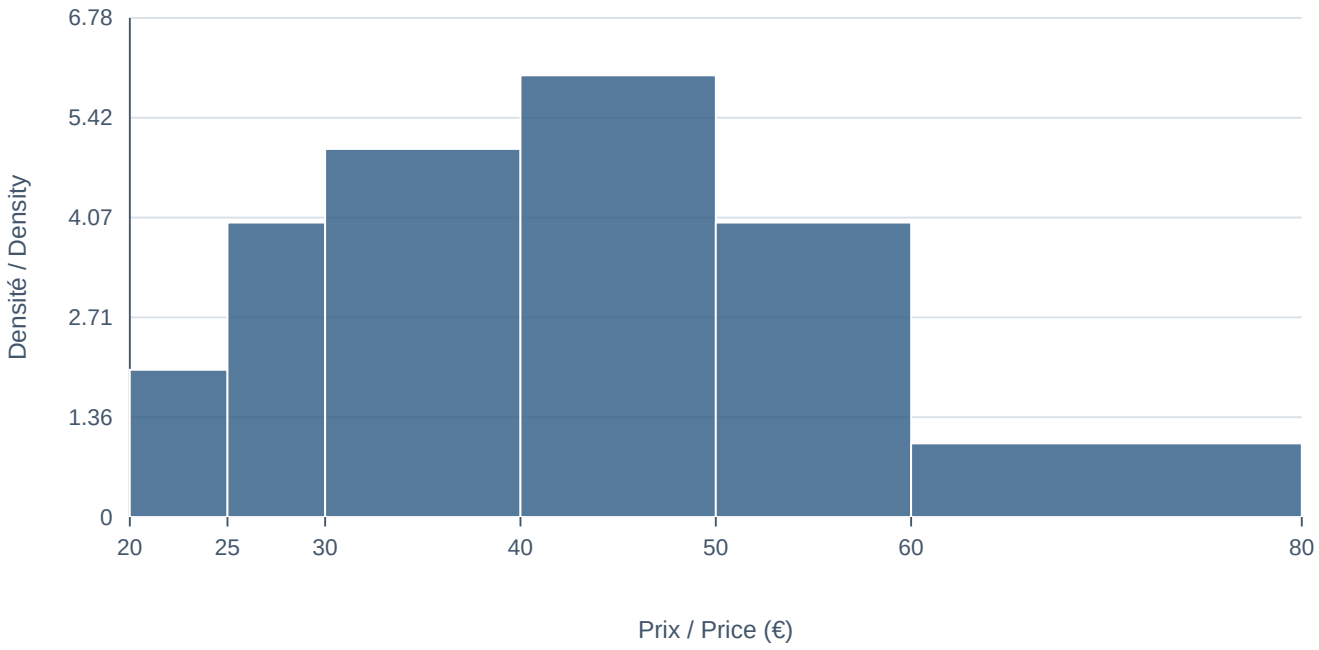
**5.** If at least 60% must be able to pay, no more than 40% may have a willingness to pay below the chosen price. Forty per cent of 200 is 80, reached at €40. The decision uses the 40th percentile, not the 60th: the recommended maximum price is therefore **€40**.

**Exam habit.** State the rule, show the calculation, then interpret the result in the context of the decision.

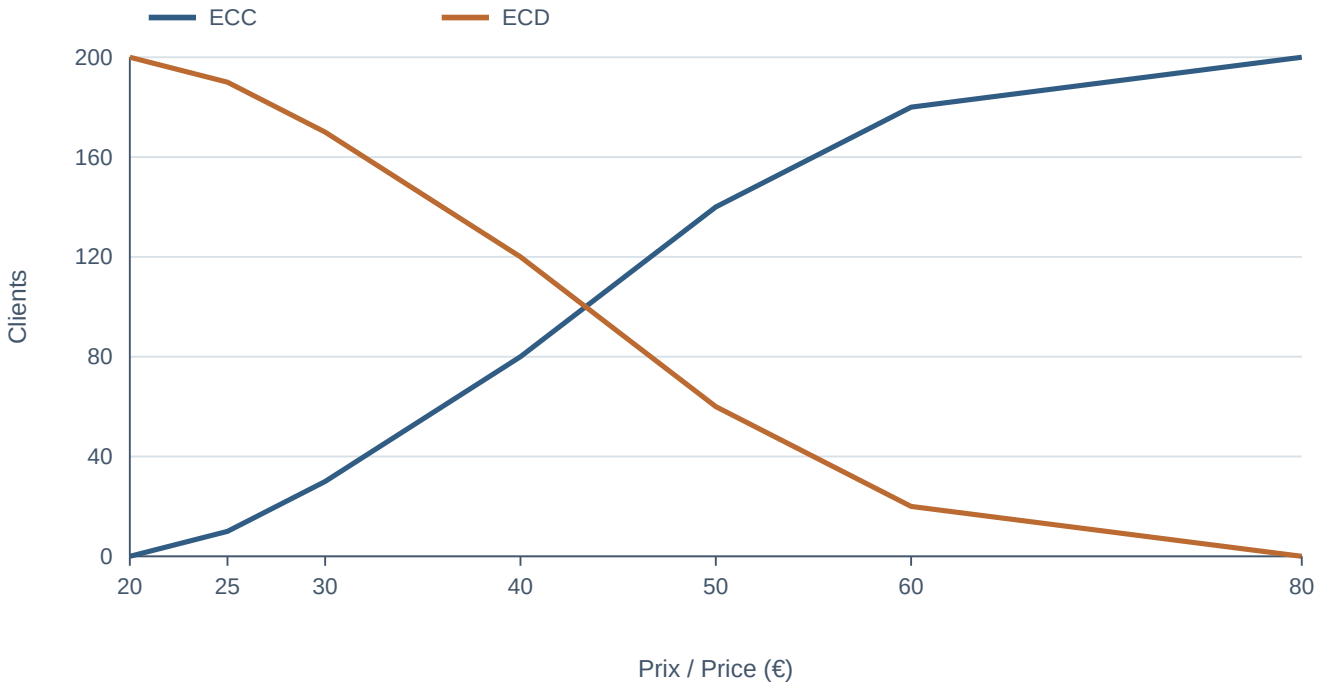
Full table and charts (histogram heights = counts/class widths):

| Classe / Class | $n_i$ | $a_i$ | $f_i$ (%) | $d_i$ | ECC | ECD | FCC (%) | FCD (%) |
|---|---|---|---|---|---|---|---|---|
| [20,25[ | 10 | 5 | 5.00 | 2 | 10 | 200 | 5.00 | 100.00 |
| [25,30[ | 20 | 5 | 10.00 | 4 | 30 | 190 | 15.00 | 95.00 |
| [30,40[ | 50 | 10 | 25.00 | 5 | 80 | 170 | 40.00 | 85.00 |
| [40,50[ | 60 | 10 | 30.00 | 6 | 140 | 120 | 70.00 | 60.00 |
| [50,60[ | 40 | 10 | 20.00 | 4 | 180 | 60 | 90.00 | 30.00 |

| Classe / Class | $n_i$ | $a_i$ | $f_i$ (%) | $d_i$ | ECC | ECD | FCC (%) | FCD (%) |
|---|---|---|---|---|---|---|---|---|
| [60,80[ | 20 | 20 | 10.00 | 1 | 200 | 20 | 100.00 | 10.00 |




Consentement à payer / Willingness-to-pay histogram




Courbes cumulatives / Cumulative counts

The €40 threshold is the 40th percentile: 80 out of 200 customers accept less than €40, so approximately 60% can afford at least €40.

Assessment / exam practice 30 min

### Extra 4 — Exam-style synthesis: reporting customer satisfaction

**Independent case — 10 marks.** Two stores collect satisfaction ratings on an ordered scale from 1 (very dissatisfied) to 5 (very satisfied).

| Rating | 1 | 2 | 3 | 4 | 5 | Total |
|---|---|---|---|---|---|---|
| **Store A** | 6 | 10 | 18 | 36 | 30 | 100 |
| **Store B** | 4 | 12 | 30 | 34 | 20 | 100 |

1. Identify the variable and its measurement scale. *(1 mark)*
2. Choose and justify one graph for comparing the two stores. Describe precisely how it should be constructed. *(2 marks)*
3. Compute, for each store, the percentages rated 4 or 5 and the cumulative percentage rated at most 2. *(3 marks)*
4. A manager says: “Store A is 50% better because it has 30 ratings of 5 versus 20.” Criticise this claim. *(2 marks)*
5. Write a three-sentence statistical conclusion, including one limitation of the survey. *(2 marks)*

Keep the order of the categories, compare percentages because the store totals could differ in another dataset, and separate description from causation.

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### Extra 4 — Exam-style synthesis: reporting customer satisfaction

#### Detailed solution

**1.** Satisfaction is qualitative ordinal: categories have a meaningful order, but the distances between adjacent ratings are not guaranteed equal.

**2.** Use grouped bars or a 100% stacked ordered bar chart. Keep ratings 1–5 in order, use identical axes and colours, and show percentages. A histogram is inappropriate because the categories are ordinal, not continuous intervals.

**3.** Store A: ratings 4–5 = (36+30)/100 = **66%**; ratings at most 2 = (6+10)/100 = **16%**. Store B: ratings 4–5 = (34+20)/100 = **54%**; ratings at most 2 = (4+12)/100 = **16%**.

**4.** Thirty is indeed 50% greater than twenty, but this isolates one category and does not mean that the store is “50% better”. Across ratings 4–5, the difference is 12 percentage points; the sampling method and uncertainty are unknown, and no causal conclusion follows.

**5. Model conclusion.** “In the surveyed samples, Store A has a higher share of favourable ratings (66%) than Store B (54%), while both have 16% ratings of 1 or 2. The difference is descriptive and does not by itself establish that Store A provides better service. Before generalising, we need to know how customers were selected, the response rates and whether the survey periods were comparable.”

## 3. Optional advanced practice

Explore additional interpretations and professional applications.

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Advanced practice  15 min

## Extra 2 — Data-quality and communication protocol

A CRM extract contains duplicate customers, missing ages coded 999, inconsistent dates and revenue recorded in euros and thousands of euros. Propose a reproducible six-step protocol before producing a management dashboard.

Separate validation, cleaning, documentation and communication.

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## Extra 2 — Data-quality and communication protocol

### Detailed solution

1. Preserve an immutable raw file and identifiers.
2. Define a data dictionary and units.
3. Detect duplicates and specify a defensible retention rule.
4. Recode 999 as missing and report missingness by group.
5. Validate dates, ranges and logical consistency.
6. Standardise monetary units, document every transformation and publish checks with the dashboard.

Never silently replace missing observations or delete unusual values.